Institute for
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E-21



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# Lecture 04

In the last lecture, we introduced the notion of a statistical model and the Fisher information matrix defined for each point of a statistical model. The aim of this lecture is to introduce a class of statistical models that is particularly important in statistics and in information geometry, the class of the so-called *exponential families*. To motivate it, we outline the maximum entropy method which was proposed by Jaynes in 1957 based on Shannon entropy (1948).

## Uncertainty and information

Consider an election with two candidates, say Alice and Bob. Before the election we have some expectation about the outcome of the election, naturally in terms of a probability vector. The vector

$$p = (0.7, 0.3),$$

for instance, encodes out expectation that Alice will win the election with probability 0.7 and Bob will win the election with probability 0.3. With this encoding of our expectation we have some uncertainty about the outcome. Which probability vector stands for the highest uncertainty? Clearly, this is the case when we expect both outcomes with the same probability, that is $p = (0.5, 0.5)$.

Which probability vector stands for a low uncertainty? This is the case when we expect one of the two candidates to be the winner already *before* the outcome of the election is known. This kind of strong belief is encoded by, for instance, $p = (0.99, 0.01)$ or $p = (0.001, 0.999)$.

Once we have a notion of uncertainty, we can define information gain as the reduction of uncertainty through the communication of the election outcome. If the uncertainty was high prior to the election then the information gain will also be high once that outcome of the election is known. If the uncertainty is low prior to the election then the information gain will also be low. This is intuitively clear because with low uncertainty, the outcome is almost known before the election so that there will be no surprise after the election.

This idea underlying the theory of information is due to C. Shannon (1948) who is considered to be the founder of information theory. The central definition of uncertainty or, equivalently, information is the following:

$$\begin{aligned} H : \mathcal{P}(I) = \mathcal{P}([n]) =: \mathcal{P}_n &\to \mathbb{R} \\ p = (p_1, ..., p_n) &\mapsto H(p) = -\sum_{i=1}^{n} p_i \ \ln(p_i). \end{aligned} \tag{1}$$

This quantity is also called *entropy*. We will refer to it as *Shannon information*. Following the above intuition with the election, we now explore the Shannon information for vectors with the highest/lowest information gain.

Consider the vector

$$p = (0, ..., 0, \underset{\substack{\uparrow \\ i^{th} \text{ position}}}{1}, 0, ..., 0).$$

At first sight, we cannot directly use (1) to evaluate the Shannon information of $p$. While $1 \cdot \ln(1) = 0$, it is not clear what $0 \cdot \ln(0)$ should mean. Definitely, $\ln(0)$ is not defined. However, $0 \cdot \ln(0)$ is actually defined in the following sense:

$$\lim_{x \searrow 0} x \ln(x) = 0 \qquad\qquad \text{(Exercise)}.$$

Thus, with the convention $0 \cdot \ln(0) = 0$, we can evaluate (1) and obtain 0 for the above vector $p$. This is consistent with our intuition. Now let us come to the vector with the highest uncertainty/information gain, that is

$$p = (\underbrace{\tfrac{1}{n}, ..., \tfrac{1}{n}}_{n \text{ times}}).$$

Does this vector maximise the Shannon information? In order to answer this question, we use the method of Langrange multipliers. This method allows us to maximise the expression (1) under the constraint

$$\sum_{i=1}^{n} p_i = 1.$$

With a Lagrange parameter $\lambda$, we consider the partial derivatives of the Lagrange function:

$$\begin{aligned}
&\frac{\partial}{\partial p_r} \left\{ -\sum_{i=1}^{n} p_i \ln(p_i) \;+\; \theta \cdot \left( \sum_{i=1}^{n} p_i - 1 \right) \right\} \\
&= -\sum_{i=1}^{n} \left( \delta_{ir} \ln(p_i) + p_i \frac{1}{p_i} \delta_{ir} \right) + \theta \\
&= -\ln(p_r) - 1 + \theta \overset{!}{=} 0 \quad \text{for all } r.
\end{aligned}$$

$\Rightarrow$

$$p_r = e^{\theta - 1}, \quad r = 1, ..., n.$$

Summation over $r$ leads to

$$1 \overset{!}{=} \sum_{r=1}^{n} p_r = \sum_{r=1}^{n} e^{\theta-1} = n \cdot e^{\theta-1}$$

$\Rightarrow$

$$\theta = \ln\left(\frac{1}{n}\right) + 1.$$

This finally leads to

$$p_r = e^{\theta-1} = e^{\left(\ln\left(\frac{1}{n}\right)+1-1\right)} = \frac{1}{n}, \quad r = 1, ..., n.$$

This confirms our intuition that the uniform probability vector should encode the expectation with maximal uncertainty.

# Maximum entropy method

Consider a probability measure $p \in \mathcal{P}_n$ and functions $f_j : I \to \mathbb{R}, \quad j = 1, ..., d$. Say that we know the mean values $\mathbb{E}_p(f_j) =: c_j, \quad j = 1, ..., d$, but have no further information about $p$. How can we infer $p$? Typically, there will be many probability measures that give us the same mean values $c_j, j = 1, ..., d$, as the true measure $p$. Their set is given by

$$\mathcal{C} = \{q \in \mathcal{P}_n : \mathbb{E}_q(f_j) = c_j, \quad j = 1, ..., d\}.$$

The maximum entropy method now suggests to select the probability measure $p^* \in \mathcal{C}$ that maximises the entropy in $\mathcal{C}$ as the best "reconstruction" of $p$.

We can find $p^*$ by the method of Lagrange multipliers. The derivation is similar to the one above, except that we now have additional constraints:

$$\begin{aligned}
&\frac{\partial}{\partial p_r}\left\{-\sum_{i=1}^n p_i \ln(p_i) + \theta\left(\sum_{i=1}^n p_i - 1\right) + \sum_{j=1}^d \theta_j \left(\sum_{i=1}^n p_i f_j(i) - c_j\right)\right\} \\
&= -\sum_{i=1}^n \left(\delta_{ir}\ln(p_i) + p_i \frac{1}{p_i}\delta_{ir}\right) + \theta + \sum_{j=1}^d \theta_j \left(\sum_{i=1}^n \delta_{ir} f_j(i)\right) \\
&= -\ln(p_r) - 1 + \theta + \sum_{j=1}^d \theta_j f_j(r) \overset{!}{=} 0, \quad r = 1, ..., n,
\end{aligned}$$

$\Rightarrow$

$$\ln(p_r) = (\theta - 1) + \sum_{j=1}^k \theta_j f_j(r), \quad r = 1, ..., n,$$

$\Rightarrow$

$$p_r = e^{(\lambda - 1)} \cdot e^{\sum_{j=1}^d \theta_j f_j(r)}, \quad r = 1, ..., n$$

The condition $\sum_{r=1}^n p_r = 1$ then leads to

$$p_r = \frac{1}{Z}\exp\left(\sum_{j=1}^d \theta_j f_j(r)\right), \tag{2}$$

where

$$Z = \sum_{r=1}^m \exp\left(\sum_{j=1}^d \theta_j f_j(r)\right).$$

Thus, the maximum entropy method yields probability measures from a statistical model of the structure (2). This motivates the following definition.

**Definition.** *(Exponential family)* Let $\mu_0$ be a reference measure in $\mathcal{M}_+(I)$, and let $f_j$, $j = 1, ..., d$, be functions on $I$. Then the statistical model

$$\begin{aligned}
\mathcal{E} &:= \mathcal{E}(\mu_0, f_1, ..., f_d) \\
&:= \left\{\frac{1}{Z}\exp\left(\sum_{j=1}^d \theta_j f_j\right)\mu_0 : \quad \theta_1, ..., \theta_d \in \mathbb{R}\right\}
\end{aligned}$$

is called an *exponential family*. Here,

$$Z = \sum_{i \in I} \exp\left(\sum_{j=1}^{d} \theta_j f_j(i)\right) \mu_0(i)$$

is called the *partition function*.

In a more general context, one considers a reference measure $\mu_0$ on a measurable space $(\Omega, \mathcal{A})$ and measurable functions $f_j : \Omega \to \mathbb{R}, \quad j = 1, ..., d$. With that one can define an exponential family in the same way.

$\Diamond$

**Examples.**

(a) The family of one-dimensional *normal distributions*:

$$\mu_0 = \lambda = \text{Lebesgue measure on } \mathbb{R}$$
$$p(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

The exponent can be decomposed as follows:

$$\begin{aligned} -\frac{(x-\mu)^2}{2\sigma^2} &= -\frac{x^2 - 2x\mu + \mu^2}{2\sigma^2} \\ &= \frac{\mu}{\sigma^2} x - \frac{1}{2\sigma^2} x^2 - \frac{\mu^2}{2\sigma^2} \\ &= \theta_1\, x + \theta_2\, x^2 - \frac{\mu^2}{2\sigma^2} \\ \Rightarrow \qquad & f_1(x) = x, \quad f_2(x) = x^2. \end{aligned}$$

Note that we can ignore the constant $-\frac{\mu^2}{2\sigma^2}$ here. This is a two-dimensional exponential family.

(b) The family of *binomial distributions*: This is given when we repeat an experiment with two outcomes $m$ times. One outcome is called "success" and the probability of success is denoted by $s$:

$$\begin{aligned} p(k; s) &= \binom{m}{k} s^k\, (1-s)^{m-k} \\ &= \binom{m}{k} e^{k \ln s}\, e^{(m-k)\ln(1-s)} \\ &= \binom{m}{k} e^{k \ln \frac{s}{1-s} + m \ln(1-s)} \end{aligned}$$

Now,

$$\mu_0(k) = \binom{m}{k}, \quad f_1(k) = k, \quad \theta_1 = \ln \frac{s}{1-s}.$$

This is a one-dimensional exponential family.

(c) The family of *Poisson distributions*:

$$\begin{aligned} p(k;s) &= \frac{\lambda^k c^{-\lambda}}{k!} \\ &= \frac{1}{k!} e^{k \ln \lambda - \lambda} \end{aligned}$$

Thus,

$$\mu_0(k) = \frac{1}{k!}, \quad f_1(k) = k, \quad \theta_1 = \ln \lambda.$$

This is again a one-dimensional exponential family.

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